Morsal Mubarak

Computational Researcher & Software Developer at the Intersection of Data Science, Neuroscience, and Public Health

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EDUCATION

Justice High School, Falls Church, Virginia · Class of 2027 — Weighted GPA 4.519 · SAT 1600

RESEARCH

Independent research study · Sole author · Under review, *Scientific Reports* — Aug 2026

- *A Reproducible Public-Data Benchmark of Birthweight-for-Gestational-Age Deviations and Infant/Fetal Mortality Risk: A Cross-Sectional Ecological Analysis of United States Vital Statistics, 2016–2024.*
- Built from **33,589,282** U.S. natality records (NCHS public-use files); code and data are public on GitHub and archived on Zenodo (DOI 10.5281/zenodo.21894038).

Traumatic brain injury (TBI) MRI computational study, Mass General Brigham / MGH Martinos Center · Lead Computational Researcher — Spring 2026

- Built a verified Python pipeline (~4,800 lines; pandas, NumPy, SciPy, statsmodels) linking MR spectroscopy to post-concussion symptoms in a **35-participant** treatment study, with outlier screening, FDR correction, and an automated re-run audit.

PoxHostAtlas · First & corresponding author, with Deon Nguyen (Arizona State University) — 2026

- Manuscript: *PoxHostAtlas: A reproducible cross-study atlas of host gene-expression remodeling during poxvirus infection.* Re-analyzes public poxvirus RNA-sequencing datasets with one common workflow; **1,834** host genes changed reproducibly across four datasets. Code: github.com/mubarsad012/poxhostatlas.

COVID-BRAIN imaging study, Athinoula A. Martinos Center (MGH) · Co-author (2nd author, manuscript) — Oct 2026

- Poster: *Lower thalamic creatine and glutathione associate with cognition and systemic inflammation in neurological post-acute sequelae of COVID-19 infection from the multicenter COVID-BRAIN imaging cohort.* Clinical Research Day, Mass General Brigham, Oct 2026.

Neurocysticercosis systematic review (ESM-2 + AlphaFold) · Senior (last) author — Manuscript in preparation

2026 Student Research Institute, Harvard Undergraduate OpenBio Laboratory · Mentee — Jun 8 – Jul 31, 2026

Bay Area Medical Explorers · Researcher — 2026

- Co-first author: *Assessing the Associations Between Antiretroviral Treatment Resource Shortage and HIV Prevalence Among Teenagers Residing in Zambia* (Mar 2026).
- First author (of 7): *Academic Pressure and Mental Health in South Asian American High School Students* (Jun 2026).

SOFTWARE

ProxyHealth AI · proxyhealth-ai.onrender.com — Public beta, Oct 2026

- Explains official papers, such as medical bills and benefits letters, in 27 languages: what each paper is, what it asks for, and a step-by-step plan, with every date and amount quoted from the paper and checked in code.
- Python/FastAPI backend; TypeScript/Expo front end (web, iOS, Android). Code: github.com/mubarsad012/ProxyHealth-AI.

LEADERSHIP & COMMUNITY

- **Emergency Afghan evacuation fundraiser** (2021): organized with father and brother; raised **$220,000+**.
- **Medical & legal interpreter** (English–Pashto): one of the main fluent English speakers in my household; interpret for family and refugee community members.
- **Public Health Youth Ambassador**, Fairfax County Health Department.
- **Co-President**, Al Shami Cultural Club · **President**, HOSA · **President**, Math Honor Society.

HONORS

- **2nd place**, PHYAP Youth Ambassador Competition Final (Georgetown University School of Medicine faculty judges), for *Nourish to Flourish: Empowering Communities through Food Education and EBT Access*, a policy proposal.

SKILLS

Programming: Python, TypeScript · **Tools:** Git, GitHub, Streamlit, LaTeX (Overleaf) · **Languages:** English, Pashto